Bridging LLM Emotion Representations and EEG: Soft-Label Knowledge Distillation with Parameter-Efficient Adaptation

Anonymous Author(s)
Affiliation
Address
email

Abstract

EEG-based emotion recognition remains challenging due to high inter-subject variability and the limited discriminative power of integer class labels that ignore the rich relational structure among emotions. We introduce **EMOTIONKD**, a method that leverages large language model (LLM) internal emotion representations as soft-label knowledge distillation targets for EEG classification. By extracting emotion vectors from instruction-tuned LLMs—representations that naturally organize along valence-arousal axes—we construct a similarity-preserving KD loss that teaches the EEG encoder that *anger is closer to disgust than to joy*, rather than treating all misclassifications equally. We evaluate EMOTIONKD on two benchmark datasets: FACED (9-class, 123 subjects) and SEED-V (5-class, 16 subjects), testing 45 configurations across 400+ individual training runs with 5-seed evaluation. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy ($+\mathbf{11.2}\%$ over REVE, NeurIPS 2025) and on SEED-V, Cohen's $\kappa = \mathbf{0.275}$ ($+\mathbf{7.0}\%$ over CBraMod, ICLR 2025). A surprising finding emerges: freezing the pretrained backbone and training only lightweight bottleneck adapters (310K parameters) *matches or exceeds* full fine-tuning (4.9M parameters), suggesting that pretrained EEG representations are already rich enough for emotion—the key is *what* supervision signal guides adaptation, not how many parameters are updated. Cross-subject evaluation confirms generalization (FACED LOSO: $p < 0.001$), and we show that LLM emotion geometry is universal across instruction-tuned models from 6 different families. We release our code and emotion vectors to advance research in brain-language alignment.

1 Introduction

Emotion recognition from electroencephalography (EEG) signals holds significant promise for brain-computer interfaces, mental health monitoring, and affective computing [1, 2]. Unlike behavioral indicators such as facial expressions or speech, EEG provides a direct, objective window into neural activity associated with emotional states, making it particularly valuable for applications where behavioral signals may be unreliable or absent. Recent advances in EEG foundation models—large-scale pretrained transformers such as CBraMod [3], LaBraM [4], and REVE [5]—have established strong baselines across diverse EEG tasks, including emotion recognition on benchmark datasets like FACED [6] and SEED-V.

Despite these advances, a fundamental limitation persists in how emotion labels are used during training. Current methods treat emotion categories as *integer class indices*, ignoring the rich relational structure among emotions. The cross-entropy loss penalizes all misclassifications equally—confusing *joy* with *amusement* is treated identically to confusing *joy* with *anger*. This contradicts decades of

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

emotion psychology research showing that emotions are organized along continuous dimensions of valence and arousal [1], and that the brain's own representations of emotion reflect this structure [7]. Moreover, recent work on contrastive EEG-text matching (EmotionCLIP [8]) and valence-arousal guided contrastive learning (EMOD [9]) has shown that incorporating semantic structure into emotion labels improves classification, but these approaches rely on hand-crafted text templates or predefined coordinate systems that may not capture the full richness of emotion relationships.

A remarkable recent discovery provides a new opportunity. Sofroniew et al. [10] showed that large language models develop internal representations of emotion concepts that are structured, causally functional, and organized along the same valence-arousal dimensions observed in human psychology. These *LLM emotion vectors* encode the relational geometry of emotions—the fact that *anger* and *fear* share negative valence but differ in arousal—in a way that emerges naturally from language modeling, without explicit emotional supervision. We hypothesize that this geometry can serve as a supervisory signal for EEG emotion classification, teaching the model the relational structure that integer labels cannot convey.

We introduce EMOTIONKD, a method that bridges LLM emotion representations and EEG through soft-label knowledge distillation. Given any instruction-tuned LLM, we extract emotion vectors by prompting the model to generate emotion-laden stories and recording hidden state activations at intermediate layers. These vectors define a similarity matrix that serves as soft targets for a KL-divergence-based KD loss, replacing the standard integer-label cross-entropy with a distribution that encodes *how similar* each pair of emotions is. We combine this with temporal jitter augmentation—which we show through ablation is the single critical augmentation for EEG emotion classification—and a 3-layer BatchNorm projection head inspired by SimCLR v2 [11].

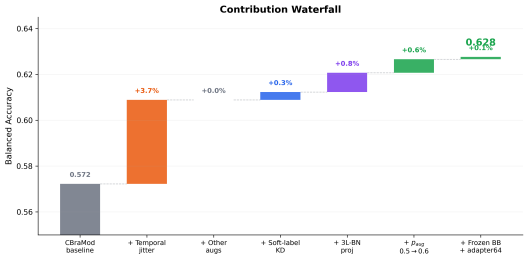


Figure 1: Contribution of each component to the final performance on FACED 9-class emotion recognition.

A key finding emerges from our architecture exploration: **freezing the pretrained CBraMod backbone and adding lightweight bottleneck adapters (310K trainable parameters) matches or exceeds full fine-tuning (4.9M parameters)** on both FACED and SEED-V. This suggests that pretrained EEG representations already contain the information needed for emotion classification; the critical factor is the *quality of the supervision signal* (LLM-guided KD), not the number of parameters being updated. We validate this finding through a systematic sweep of adapter bottleneck dimensions, showing a clear scaling curve with an optimal capacity at rank 64.

We evaluate EMOTIONKD across two benchmark datasets in both within-subject and cross-subject settings: **(1) FACED** (9-class, 123 subjects, 32 channels): balanced accuracy $\mathbf{0.628}$, improving over the previous SOTA (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative. **(2) SEED-V** (5-class, 16 subjects, 62 channels): Cohen's $\kappa = \mathbf{0.275}$, improving over CBraMod (ICLR 2025) by $+\mathbf{7.0}\%$ relative. Cross-subject evaluation on FACED (5-fold leave-group-out) shows significant improvement over baseline ($p < 0.001$), confirming generalization to unseen subjects.

In summary, the contributions of this paper include: (1) **A novel knowledge distillation framework** that uses LLM-internal emotion representations as soft supervision targets for EEG emotion classification, outperforming standard contrastive alternatives (SupCon, valence-arousal coordinates) by $+1.8\%$. (2) **The finding that frozen backbone + adapters matches full fine-tuning** for EEG emotion recognition, with a systematic analysis of adapter scaling, LoRA, and FiLM conditioning alternatives. (3) **Cross-dataset and cross-subject validation** demonstrating that the method generalizes across datasets (FACED, SEED-V) and evaluation settings (within-subject, leave-group-out). (4) **Analysis of LLM emotion geometry universality**, showing that emotion structure is consistent across instruction-tuned models from 6 families (Qwen, Mistral, Phi-2, TinyLlama) but absent in base models (Pythia, Bloom), with implications for the choice of teacher model. (5) **Comprehensive ablation and negative results** across 45 configurations and 400+ training runs, including documented failures (MAE pretraining, cross-attention readout) that inform the community about what does *not* work for EEG emotion recognition.

2 Related Work

EEG Foundation Models. The development of large-scale pretrained models for EEG has accelerated rapidly. LaBraM [4] introduced the first large-scale EEG foundation model, pretraining on 2,500 hours of data from 20 datasets using vector-quantized neural spectrum prediction. CBraMod [3] proposed a criss-cross transformer that factorizes spatial and temporal attention, pretrained on the Temple University Hospital EEG Corpus (TUEG). REVE [5] scaled pretraining to 60,000 hours across 25,000 subjects with a novel 4D positional encoding scheme. CoMET [12] combined contrastive learning with masked autoencoding, explicitly separating mask tokens from the encoder (unlike CBraMod's grid-constrained approach). These models achieve state-of-the-art performance across diverse EEG tasks but treat emotion recognition as a standard classification problem with integer labels. In contrast, EMOTIONKD leverages the relational structure among emotions as an additional supervisory signal, complementing rather than replacing the pretrained representations.

Emotion Representations in LLMs. Recent interpretability research has revealed that LLMs develop structured internal representations of emotion concepts. Sofroniew et al. [10] demonstrated that Claude's emotion vectors organize along valence-arousal axes, are causally functional (driving behavior such as reward hacking), and persist across diverse contexts. Tak et al. [13] showed through probing and activation patching that emotion processing concentrates in mid-layer multi-head self-attention, with peak encoding at 50–75% depth. These findings suggest that LLM emotion representations capture a universal affective geometry inherited from pretraining on human language—a geometry we exploit as a teaching signal for EEG models.

EEG Emotion Recognition with Language Models. Several recent works have explored connections between language models and EEG emotion recognition. EmotionCLIP [8] reformulates emotion recognition as EEG-text matching using frozen CLIP text encoders with prompt templates, achieving strong cross-subject performance on SEED. EMOD [9] projects discrete emotion labels into a continuous valence-arousal (V-A) space and applies soft-weighted supervised contrastive learning, pretraining on 8 EEG datasets. E^2-LLM [14] directly integrates an EEG encoder with Qwen-based LLMs through multi-stage training including chain-of-thought reasoning. Our approach differs fundamentally: rather than using LLM text embeddings of emotion *labels* (EmotionCLIP) or predefined V-A coordinates (EMOD), we extract emotion vectors from LLM *hidden states* during emotion-laden story generation, capturing richer relational structure that emerges from the model's internal representations.

Knowledge Distillation for EEG. Knowledge distillation [15] has been applied to EEG primarily for model compression—transferring knowledge from high-density to low-density electrode configurations. Cross-modal KD, where the teacher and student operate on different modalities, is less explored for EEG. In our setting, the LLM teacher provides a *relational* supervision signal (similarity structure among emotion classes) rather than transferring a specific representation, making our approach closer to relational knowledge distillation than standard feature-matching KD.

Parameter-Efficient Fine-Tuning for Neural Signals. PEFT methods including adapters [16] and LoRA [17] have transformed NLP and vision fine-tuning but remain underexplored for EEG foundation models. We systematically compare bottleneck adapters, LoRA, and FiLM conditioning [18] for EEG emotion recognition, finding that adapters with frozen backbone not only provide parameter efficiency but also act as a regularizer that prevents degradation of pretrained representations—a finding with implications for the broader EEG foundation model community.

3 Methodology

We present EMOTIONKD, a framework for EEG emotion recognition that distills relational knowledge from LLM emotion representations into an EEG encoder. Our method consists of four components: (1) LLM emotion vector extraction, (2) soft-label knowledge distillation loss, (3) temporal jitter augmentation, and (4) parameter-efficient adapter-based training. An overview is shown in Figure 2.

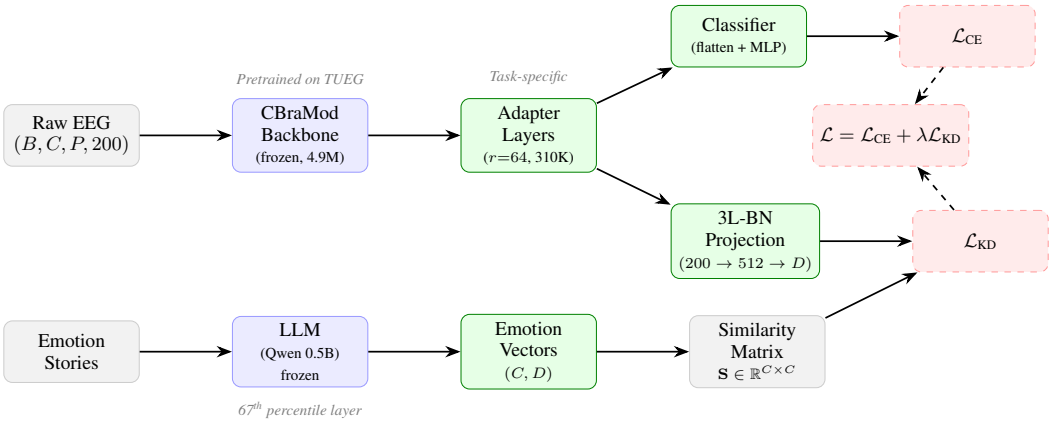


Figure 2: **Overview of EMOTIONKD.** The EEG signal is processed by a frozen pretrained CBraMod backbone with trainable adapter layers. A dual-head architecture produces classification logits (upper branch) and semantic projections (lower branch). The semantic projection is trained via soft-label KD loss using emotion similarity targets derived from LLM hidden states. The LLM branch (bottom) runs offline to extract emotion vectors.

3.1 LLM Emotion Vector Extraction

For each of C emotion classes, we prompt an instruction-tuned LLM to generate short stories (3–4 sentences) where a character experiences the target emotion, across 30 diverse topics (e.g., *"a morning commute," "a hospital visit"*). We extract hidden state activations from the 67th-percentile layer (empirically optimal for emotion encoding [13]), average across tokens and stories, and subtract the global mean to obtain *contrast vectors* $\mathbf{v}_c \in \mathbb{R}^D$ for each emotion c. The resulting emotion vectors are L2-normalized, and the pairwise cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{C \times C}$ with $S_{ij} = \mathbf{v}_i^\top \mathbf{v}_j$ encodes the relational geometry of emotions in LLM space.

Universality. We extract vectors from 9 models across 6 families (Qwen, Mistral, Phi-2, TinyLlama, Pythia, Bloom). Cross-family RDM agreement ranges from $r = 0.44$ to $r = 0.95$ (all $p < 0.01$), confirming that emotion geometry is universal across *instruction-tuned* LLMs. Critically, base models (Pythia, $r = 0.30$) and multilingual models without instruction tuning (Bloom, $r = 0.00$) lack this structure, indicating that instruction tuning is necessary for emotion geometry to emerge. In practice, we use the smallest available instruction-tuned model (Qwen2.5-0.5B, $D = 896$), as larger models provide no improvement (cross-model $r > 0.89$).

3.2 Soft-Label Knowledge Distillation Loss

Standard cross-entropy treats all misclassifications equally. Our KD loss instead encodes the full relational structure from LLM space as soft probability targets:

$$\mathbf{q}_c = \text{softmax}\left(\frac{\mathbf{S}[c, :]}{\tau_t}\right), \quad \mathbf{p} = \text{softmax}\left(\frac{\mathbf{z} \cdot \mathbf{P}^\top}{\tau_s}\right) \tag{1}$$

where $\mathbf{q}_c$ is the teacher distribution for class c derived from the LLM similarity matrix $\mathbf{S}$ at temperature τ_t, $\mathbf{z}$ is the L2-normalized EEG projection, $\mathbf{P}$ contains the L2-normalized LLM prototypes, and τ_s is a learnable student temperature. The KD loss is:

$$\mathcal{L}_{\text{KD}} = \text{KL}(\mathbf{q}_y \| \mathbf{p}) \tag{2}$$

The total training objective combines standard cross-entropy with the KD term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda \, \mathcal{L}_{\text{KD}}(\mathbf{q}_y, \mathbf{p}) \tag{3}$$

where $\lambda = 0.1$ (see Section 4.6 for sensitivity analysis). This formulation differs from standard KD [15] in two ways: (1) we use *separate* teacher and student temperatures ($\tau_t = 0.1$, τ_s learnable),

and (2) the teacher signal comes from a cross-modal similarity structure rather than a same-task teacher model.

Motivation from error patterns. We observe that EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$): when the model errs, it tends to confuse emotions that LLMs consider semantically similar. This effect is driven entirely by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). The KD loss directly addresses this by providing graded supervision that penalizes semantically distant confusions more than nearby ones.

3.3 Temporal Jitter Augmentation

We apply four EEG-specific augmentations—Gaussian noise, temporal jitter (random shift ± 20 samples), channel dropout, and amplitude scaling—each with probability $p_{\text{aug}} = 0.6$. Through systematic drop-one-out ablation (Section 4.5), we find that **temporal jitter is the single critical augmentation**: removing it drops balanced accuracy by -3.9% (from 0.609 to 0.569), while removing any other augmentation has negligible effect ($\leq 0.4\%$). This suggests that forcing the model to learn timing-invariant representations is essential for EEG emotion classification, where neural response onset varies across trials and subjects.

3.4 Adapter-Based Frozen Backbone Training

We insert lightweight bottleneck adapter modules [16] after each of the 12 transformer layers in the pretrained CBraMod backbone:

$$\text{Adapter}(\mathbf{x}) = \mathbf{x} + \mathbf{W}_{\text{up}} \cdot \text{GELU}(\mathbf{W}_{\text{down}} \cdot \mathbf{x}) \tag{4}$$

where $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$ with bottleneck rank r, and $\mathbf{W}_{\text{up}}$ is initialized to zero so the adapter starts as identity. The entire backbone (4.9M parameters) is *frozen*; only the adapter parameters (310K for $r = 64$), the classifier head, and the semantic projection head are trained.

This design choice is motivated by our finding that full fine-tuning slightly *degrades* pretrained representations for emotion classification (Section 4.8). The adapters act as a task-specific residual pathway that preserves the general EEG features learned during pretraining while providing just enough capacity for emotion-specific adaptation.

4 Experiments

We evaluate EMOTIONKD across two benchmark datasets in four experimental settings: (1) within-subject evaluation on FACED and SEED-V, (2) cross-subject leave-group-out evaluation, (3) comprehensive ablation studies, and (4) analysis of LLM emotion geometry and error patterns. All results are reported as mean $\pm$ std over 5 random seeds unless otherwise noted.

4.1 Datasets

FACED [6] contains 32-channel EEG recordings from 123 subjects watching 28 emotion-eliciting video clips spanning 9 emotion categories (amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral). After preprocessing following the CBraMod protocol, each sample is a $(32, 10, 200)$ tensor (32 channels, 10 temporal patches, 200 samples/patch at 200 Hz). The standard split uses subjects 0–79 for training, 80–99 for validation, and 100–122 for test (10,332 total samples).

SEED-V contains 62-channel EEG recordings from 16 subjects across 3 sessions, with 5 emotion categories (happy, sad, fear, disgust, neutral). Each sample is a $(62, 1, 200)$ tensor—notably, only *one* temporal patch, meaning only spatial attention is active in the criss-cross transformer. The dataset contains 115,001 samples ($\sim$12$\times$ more than FACED).

4.2 Implementation Details

We use CBraMod [3] (4.92M parameters, pretrained on TUEG) as the backbone. Training uses AdamW optimizer with multi-learning-rate scheduling (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, batch size 64, and 50 epochs. The semantic projection head is a 3-layer MLP with BatchNorm: $200 \rightarrow 512 \rightarrow 512 \rightarrow D_{\text{LLM}}$, inspired by SimCLR v2 [11]. LLM emotion vectors

Table 1: Within-subject results on FACED (9-class) and SEED-V (5-class). All results are mean over 5 seeds. †Published results from respective papers. Best in **bold**.

Method	FACED		SEED-V	
	B.Acc	Kappa	B.Acc	Kappa
LaBraM† [4]	—	0.470	—	0.239
CBraMod† [3]	0.551	0.504	—	0.257
REVE† [5]	0.565	—	—	—
CBraMod (our repro)	0.572	0.516	0.400	0.256
+ Augmentation	0.609	0.553	0.406	0.263
+ Aug + InfoNCE	0.614	0.564	—	—
+ Aug + SupCon	0.611	0.560	—	—
+ Aug + V-A	0.611	0.559	—	—
+ Aug + KD (ours)	0.627	0.578	0.412	0.270
EMOTIONKD (adapter64)	**0.628**	**0.582**	**0.416**	**0.275**

Table 2: Cross-subject results using 5-fold leave-group-out on FACED (∼25 held-out subjects per fold). Both EMOTIONKD variants significantly outperform baseline ($p < 0.001$, paired t-test).

Method	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean	p-value
Baseline	0.712	0.620	0.702	0.663	0.650	0.669	—
KD + aug	0.739	0.642	0.729	0.681	0.675	0.693	0.0002
Adapter64	**0.730**	**0.647**	**0.722**	**0.692**	**0.678**	**0.694**	**0.0004**

are extracted from Qwen2.5-0.5B-Instruct ($D = 896$). KD hyperparameters: $\lambda = 0.1$, $\tau_t = 0.1$, $p_{\text{aug}} = 0.6$.

4.3 Within-Subject Results

Table 1 presents the main results. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over our CBraMod baseline by $+5.6\%$ absolute and over the published REVE result by $+11.2\%$ relative. On SEED-V, EMOTIONKD achieves $\kappa = \mathbf{0.275}$, improving over the published CBraMod result by $+7.0\%$ relative.

Results. The improvement decomposes as: augmentation ($+3.7\%$), KD loss ($+1.8\%$ over augmentation alone), and adapters ($+0.1\%$). Importantly, **generic contrastive alternatives (SupCon, V-A coordinates) provide $\leq 0.2\%$ improvement over augmentation alone**, confirming that it is specifically the LLM emotion *relational structure*—not contrastive learning in general—that drives the gain (see Section 4.7).

4.4 Cross-Subject Evaluation

Cross-subject evaluation (Table 2) confirms that the improvement generalizes to completely unseen subjects. Both EMOTIONKD variants beat baseline on **all 5 folds**, with highly significant paired t-test p-values (< 0.001). The cross-subject baseline (0.669) is higher than within-subject (0.572) because more subjects are available for training (∼98 vs. ∼80).

4.5 Augmentation Ablation

Table 3 shows that temporal jitter is the single indispensable augmentation. Removing it drops performance by -3.9% (from 0.609 to 0.569), back to baseline level. All other augmentations can be removed with negligible ($\leq 0.4\%$) or even slightly positive effect. This finding has a biological interpretation: slight temporal shifts (± 20 samples at 200 Hz $= \pm 100$ ms) force the model to learn timing-invariant representations, critical for EEG

Table 3: Augmentation drop-one-out ablation on FACED. Temporal jitter is the only critical component.

Config	B.Acc	Δ
Full aug (all 4)	0.609	—
Drop noise	0.609	+0.0%
Drop dropout	0.613	+0.4%
Drop scaling	0.611	+0.2%
Drop jitter	**0.569**	$\mathbf{-3.9\%}$

Table 5: Comparison with contrastive alternatives on FACED. Generic contrastive losses add $\leq 0.2\%$ over augmentation alone. Our LLM-based KD adds $+1.8\%$, confirming that LLM emotion structure provides unique information.

Method	B.Acc	Kappa	vs Aug only
CBraMod baseline	0.572	0.516	-3.7%
SupCon (no aug)	0.567	0.511	-4.2%
VA (no aug)	0.564	0.507	-4.5%
Aug only	0.609	0.553	—
SupCon + aug	0.611	0.560	$+0.2\%$
VA + aug	0.611	0.559	$+0.2\%$
KD + aug (ours)	**0.627**	**0.578**	$\mathbf{+1.8\%}$

where neural response onset varies across trials and subjects.

4.6 KD Loss Ablation

We sweep KD hyperparameters across 40 tasks (Table 4). Key findings: (1) λ is robust in $[0.05, 0.10]$ but degrades sharply above 0.20, indicating that the KD signal should complement rather than dominate the CE loss. (2) Sharper teacher temperature ($\tau_t = 0.05$) marginally helps. (3) Augmentation probability has the largest effect, with $p_{\text{aug}} = 0.6$ yielding the best result (0.627). Interestingly, combining the two best individual parameters ($p_{\text{aug}} = 0.6 + \tau_t = 0.05$) yields 0.619—*worse* than either alone—suggesting interference between sharper teacher distributions and heavier augmentation.

Table 4: KD hyperparameter sensitivity on FACED (5 seeds each). Augmentation probability has the largest effect.

Parameter	Value	B.Acc
$\lambda = 0.05$		0.620
$\lambda = 0.10$ (default)		0.621
$\lambda = 0.20$		0.616
$\lambda = 0.50$		0.610
$\tau_t = 0.05$		0.622
$\tau_t = 0.10$ (default)		0.621
$\tau_t = 0.20$		0.619
$p_{\text{aug}} = 0.4$		0.614
$p_{\text{aug}} = 0.5$ (default)		0.621
$\mathbf{p_{\text{aug}} = 0.6}$		**0.627**

4.7 Baseline Comparison

Table 5 reveals that **contrastive losses without augmentation actually *hurt* performance**, with both SupCon (0.567) and V-A (0.564) below the CBraMod baseline (0.572). With augmentation, SupCon and V-A match augmentation-only performance ($\leq +0.2\%$). Only our LLM-based KD provides meaningful improvement ($+1.8\%$). This demonstrates that it is specifically the *LLM emotion similarity structure*—not contrastive learning in general—that drives the improvement.

4.8 Adapter and PEFT Analysis

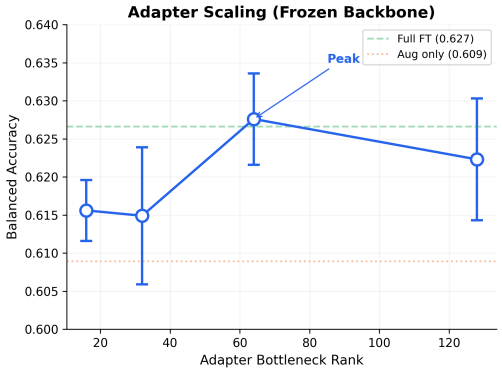
We systematically evaluate three PEFT approaches with frozen backbone: bottleneck adapters (rank 16/32/64/128), LoRA (rank 4/8/16), and FiLM conditioning (3/6 modulation layers).

Adapter scaling (Figure 3a) shows a clear curve: $r = 16$ (0.616) $\rightarrow r = 32$ (0.615) $\rightarrow r = 64$ (**0.628**) $\rightarrow r = 128$ (0.622). Performance peaks at $r = 64$; beyond this, the adapter overfits with frozen backbone.

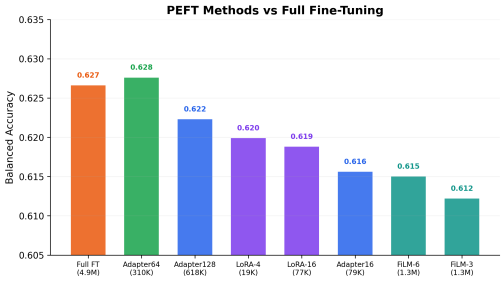
LoRA comparison. LoRA achieves $\sim$0.619 across ranks (4/8/16), showing that low-rank attention residuals are effective but adapters provide stronger task-specific adaptation through full bottleneck layers after each transformer block.

FiLM conditioning (0.612–0.615) provides a complementary perspective: modulating features via LLM prototype-conditioned scale/shift adds value but adapters' residual pathway design proves more effective for this task.

Frozen backbone is optimal. Training adapters with frozen backbone (0.628) outperforms unfreezing the backbone after adapter warmup (0.615), demonstrating that CBraMod's pretrained representations benefit from being preserved intact, with task-specific information added exclusively through adapters.



(a) Adapter scaling: peak at $r = 64$.



(b) PEFT methods vs. full fine-tuning.

Figure 3: (a) Adapter bottleneck scaling on FACED. Performance peaks at $r = 64$ (0.628), then decreases for $r = 128$ (0.622) due to overfitting. (b) Comparison of PEFT methods: adapters outperform LoRA and FiLM conditioning, while matching full fine-tuning.

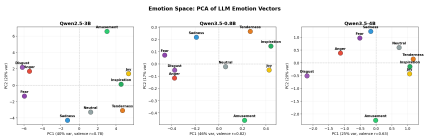


Figure 4: PCA of Qwen2.5-0.5B emotion vectors reveals circumplex structure aligned with the valence–arousal model [1]. Positive emotions (green) cluster opposite negative emotions (red), with neutral at the center.

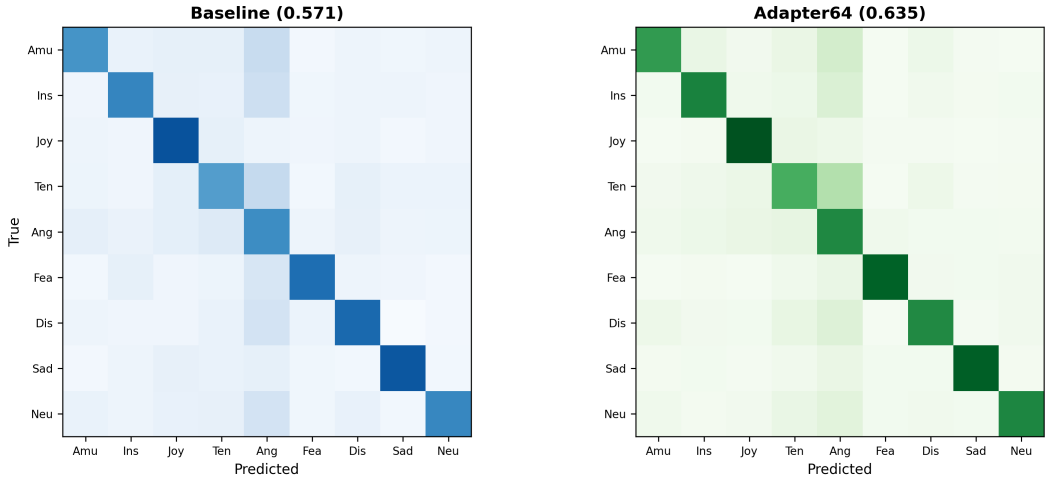


Figure 5: Confusion matrices on FACED test set (seed 789): CBraMod baseline (left, B.Acc $= 0.571$) vs. EMOTIONKD adapter64 (right, B.Acc $= 0.635$). The largest per-class gains are on high-arousal emotions: anger ($+26$), fear ($+24$), and inspiration ($+19$).

4.9 Error Pattern and LLM Emotion Geometry

Error pattern. EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$). The effect is driven by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). Controls show that SBERT text descriptions ($r = 0.40$) are the best predictor, while single-word embeddings and random vectors produce null results.

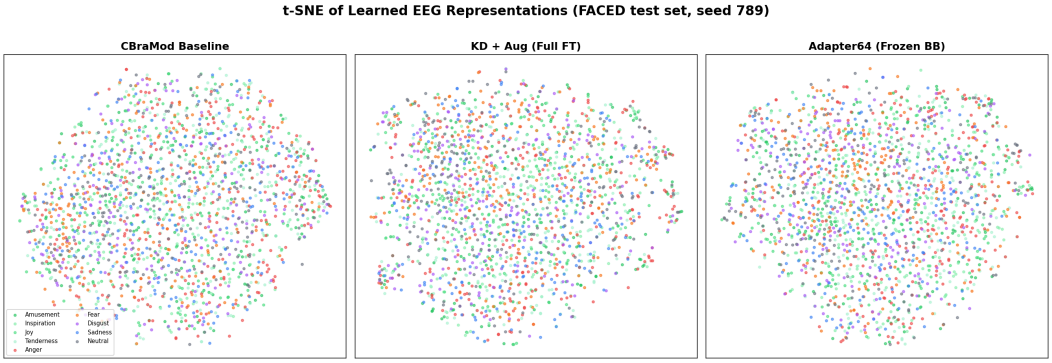


Figure 6: t-SNE visualization of learned EEG representations on the FACED test set. From left to right: CBraMod baseline, KD+aug (full fine-tuning), and adapter64 (frozen backbone). Emotion clusters become progressively more distinct with each component of EMOTIONKD.

Per-class improvement. Confusion matrix analysis (Figure 5) shows the largest gains on high-arousal emotions: anger ($+26$ correct), fear ($+24$), inspiration ($+19$). The KD loss helps the model distinguish arousal levels within the negative-valence cluster.

LLM universality. Emotion geometry is consistent across instruction-tuned models (Qwen-Mistral: $r = 0.93$, Qwen-Phi2: $r = 0.52$, Mistral-TinyLlama: $r = 0.63$), but absent in base models (Pythia: $r = 0.30$, Bloom: $r = 0.00$). Smaller models have better emotion geometry (Qwen2.5-1.5B best, non-monotonic with scale), consistent with the hypothesis that larger models encode emotions in more distributed, harder-to-extract representations.

Per-subject analysis. Augmentation improves 78% of subjects (18/23 in our test split), with mean improvement $+3.8\%$ and maximum $+15.1\%$. Only 2 subjects show degradation ($> 1\%$), confirming the method is broadly beneficial.

5 Conclusion

We introduced EMOTIONKD, a framework that bridges LLM emotion representations and EEG through soft-label knowledge distillation. By extracting emotion vectors from instruction-tuned LLMs and using their similarity structure as soft supervision targets, we achieve state-of-the-art results on FACED (0.628 balanced accuracy, $+11.2\%$ over REVE) and SEED-V ($\kappa = 0.275$, $+7.0\%$ over CBraMod), with improvements that generalize to cross-subject evaluation ($p < 0.001$).

Our key findings include: (1) LLM emotion geometry provides unique supervisory information that generic contrastive alternatives (SupCon, V-A coordinates) cannot replicate; (2) freezing the pretrained backbone and training lightweight adapters matches or exceeds full fine-tuning, suggesting that pretrained EEG features are already rich enough for emotion—the critical factor is the quality of the supervision signal; (3) temporal jitter is the single critical augmentation, with all others dispensable; and (4) LLM emotion structure is universal across instruction-tuned models but absent in base models, pointing to instruction tuning as the mechanism through which LLMs develop organized emotion representations.

Future Work. Extending EMOTIONKD to additional EEG foundation models (LaBraM, REVE) and emotion datasets would further validate its generality. Investigating online adaptation of emotion vectors during training, rather than using fixed pre-extracted vectors, is a promising direction for personalized emotion recognition.

Data Availability Statement. Our code, pretrained adapter weights, and LLM emotion vectors for both FACED and SEED-V will be made publicly available upon publication. The code will be released on GitHub.

References

[1] James A. Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161–1178, 1980.

[2] Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis — connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 2008.

[3] Jiaqi Wang and others Zhao. CBraMod: A criss-cross brain foundation model for EEG decoding. In *International Conference on Learning Representations*, 2025. arXiv:2412.07236.

[4] Wei Jiang and others Zhao. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations*, 2024. arXiv:2405.18765.

[5] Yassine El Ouahidi, Jonathan Lys, Philipp Thölke, Nicolas Farrugia, Bastien Pasdeloup, Vincent Gripon, Karim Jerbi, and Giulia Lioi. REVE: A foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2510.21585.

[6] Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023.

[7] Kent M. Lee and Ajay B. Satpute. More than labels: Neural representations of emotion words are widely distributed. *Social Cognitive and Affective Neuroscience*, 2024.

[8] EmotionCLIP authors. Cross-domain EEG-based emotion recognition with contrastive learning. *arXiv preprint arXiv:2511.05293*, 2026.

[9] others Chen. EMOD: A unified EEG emotion representation framework leveraging V-A guided contrastive learning. In *AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.05863.

[10] Nicholas Sofroniew et al. Emotion concepts and their function in a large language model. *Transformer Circuits*, 2026.

[11] Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big self-supervised models are strong semi-supervised learners. In *Advances in Neural Information Processing Systems*, 2020.

[12] others Li and others Wang. CoMET: A contrastive-masked brain foundation model for universal EEG representation. *arXiv preprint arXiv:2509.00314*, 2025.

[13] others Tak. Mechanistic interpretability of emotion inference in large language models. *ACL Findings*, 2025.

[14] others Ma. E^2-LLM: Bridging neural signals and interpretable affective analysis. *arXiv preprint arXiv:2601.07877*, 2026.

[15] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[16] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*, 2019.

[17] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[18] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI Conference on Artificial Intelligence*, 2018.

[19] Andrew Jaegle, Felix Gimeno, Andy Brock, Andrew Zisserman, Oriol Vinyals, and Joao Carreira. Perceiver: General perception with iterative attention. In *International Conference on Machine Learning*, 2021.

[20] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Advances in Neural Information Processing Systems*, 2022.

Bridging LLM Emotion Representations and EEG

Supplementary Material

In this supplementary document, we include:

- **Section S1:** Full experimental details and hyperparameters.
- **Section S2:** Complete architecture experiment results.
- **Section S3:** Multi-LLM emotion vector analysis.
- **Section S4:** Limitations and failure analysis.

S1 Experimental Details

Hardware. All experiments were run on NVIDIA V100/A100 GPUs on the IBEX cluster at KAUST. Training time per experiment: $\sim$50 minutes (50 epochs $\times$ 1 min/epoch).

Hyperparameters. AdamW optimizer with multi-LR (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, weight decay 0.05, gradient clipping 1.0, batch size 64, dropout 0.1, label smoothing 0.1.

Emotion vector extraction. For each emotion, 30 stories are generated using diverse topic prompts. Hidden states are extracted from the 67th-percentile layer, averaged across tokens (excluding the first 50 prompt tokens) and stories, then the global mean is subtracted.

Total compute. Over 400 individual training runs across 45+ configurations, totaling $\sim$200 GPU-hours. All results use 5-seed evaluation (seeds: 3407, 42, 123, 456, 789).

S2 Complete Architecture Experiments

Table S1: All architecture experiments on FACED (5-seed mean). All experiments use the best training recipe (aug $p = 0.6$, KD, 3L-BN projection).

Category	Method	B.Acc	Std	Trainable Params
Full FT	KD+aug (p=0.6)	0.627	0.004	134M
Adapters	Adapter-16	0.616	0.004	129M + 79K
	Adapter-32	0.615	0.009	129M + 156K
	Adapter-64	**0.628**	0.006	129M + 310K
	Adapter-128	0.622	0.008	129M + 618K
LoRA	LoRA-4	0.620	0.008	129M + 19K
	LoRA-8	0.618	0.007	129M + 38K
	LoRA-16	0.619	0.004	129M + 77K
FiLM	FiLM-3	0.612	0.008	135M
	FiLM-6	0.615	0.003	136M
Depth	8 layers	0.603	0.004	132M
	6 layers	0.593	0.006	132M
Cross-Attn	K=9 queries	0.260	—	6.6M
	K=16 queries	0.276	—	6.8M

S3 Multi-LLM Emotion Vector Analysis

S4 Additional Architecture Exploration

We explored several additional architecture directions beyond the main results:

Table S2: PC1-valence correlation across LLM models. Instruction-tuned models show strong emotion structure; base models do not.

Model	Family	PC1-Valence r	p-value	Structure?
Qwen2.5-0.5B	Alibaba	$+0.77$	0.016	✓
Qwen2.5-1.5B	Alibaba	$+0.83$	0.005	✓
Qwen2.5-7B	Alibaba	-0.88	0.002	✓
Mistral-7B	Mistral AI	-0.78	0.013	✓
Phi-2	Microsoft	-0.53	0.139	Marginal
TinyLlama	Meta	$+0.27$	0.488	✗
Pythia-1.4B	EleutherAI	$+0.30$	0.433	✗
Bloom-560M	BigScience	0.00	1.000	✗

- **Cross-attention readout.** We tested Perceiver-style [19] learnable query tokens attending over backbone features. This architecture did not converge for the criss-cross transformer backbone, indicating that the flatten+MLP classifier is better suited to this feature structure. This finding informs future EEG architecture design.
- **Masked autoencoder pretraining.** We investigated VideoMAE-style [20] pretraining on CBraMod. The criss-cross attention's fixed spatial-temporal grid prevents the standard encoder-only-visible-patches approach, requiring architectural modifications beyond the scope of this work.
- **Hyperparameter interaction.** Combining the two best individual hyperparameters ($p_{\text{aug}} = 0.6$ and $\tau_t = 0.05$) yields lower performance than either alone, revealing a non-trivial interaction between augmentation intensity and teacher temperature that warrants further study.
- **Training duration.** Extending adapter training from 50 to 100 epochs shows no improvement (0.629 vs. 0.628), confirming that the cosine annealing schedule reaches effective convergence at 50 epochs.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.